S13 M2 Analytical Model Contract

Source-bounded continuous mass, mandatory transfer, and finite receiving

Methodology review artifact

30 September 2026

1 Purpose and claim boundary

This contract instantiates one minimal member of the M2 family admitted by S12. It is designed to test whether hidden lane allocation can change a useful common lane-uniform variable-speed-limit (VSL) decision through endogenous mandatory transfer and finite receiving. It is not a calibrated freeway model, a paper reproduction, or an implementation specification.

The contract preserves continuous vehicle-equivalent mass, fixed lane topology, one shared infrastructure action, exact predecision aggregate-history equality, and complete affected-cohort time accounting. The analytical certificate that uses this contract is a conditional existence result.

2 Geometry, cohort, hidden state, and observer

Consider a permanent three-to-two lane drop. For the analytical reduction, the ending lane is explicit and the two continuing lanes are represented as one symmetric continuing group. Let a common affected cohort have total mass $M > 0$ veh-eq. A hidden preparation state s is described by its ending-lane fraction $p_s \in (0, 1)$:

$$e_s(0) = p_s M, \qquad c_s(0) = (1 - p_s)M. \tag{1}$$

The compared states have different p_s but the same M, route population, geometry, future forcing, action set, and declared aggregate predecision history.

The strong observer contains spatial aggregate density, flow, speed, cumulative counts, and their complete declared history. It does not contain lane decomposition. S05 supplies the conditional upstream construction in which conservative lateral transfers cancel after aggregation, so two values $p_L < p_H$ can arrive at the decision boundary with identical observer history. The decision boundary is the first entry into the fixed spatial M2 preparation zone; M2 physics is common thereafter.

3 Action and traversal time

The only action is a constant lane-uniform advice speed $v \in [v_{\min}, v_{\max}]$, with $0 < v_{\min} < v_{\max}$. The preparation zone has common length $L > 0$ km. On the selected uncongested longitudinal

branch, advice determines zone traversal time

$$\tau(v) = \frac{L}{v}, \qquad \tau \in [\tau_{\min}, \tau_{\max}] = \left[\frac{L}{v_{\max}}, \frac{L}{v_{\min}}\right]. \tag{2}$$

This is the full one-parameter action family. Fastest advice $v_{\max}$ is the endpoint $\tau_{\min}$; it is not omitted from the comparison.

4 Conservative mandatory transfer

Within the preparation zone, ending-lane mass has mandatory lateral demand. On the fixed positive active branch used for this certificate, a common transfer rate $\lambda > 0$ (h^{-1}) gives

$$\dot{e}_s(t) = -\lambda e_s(t), \qquad \dot{c}_s(t) = +\lambda e_s(t), \qquad 0 \le t \le \tau. \tag{3}$$

Thus

$$e_s(\tau) = p_s M e^{-\lambda\tau}, \qquad c_s(\tau) = M - e_s(\tau), \qquad e_s(t) + c_s(t) = M. \tag{4}$$

Equation (3) is a deterministic continuous-mass specialization of repeated mandatory CTM transfer on one active density/geometry branch. It is an explicit S13 analytical assumption. It does not assert an empirically calibrated exponential lane-change law.

Target receiving in the preparation zone is assumed nonbinding for the admitted certificate regime. This does not remove receiving from M2: the consequential finite-receiving interaction occurs at the choke. It isolates the mechanism by which longer residence can complete more mandatory transfer before that interaction.

5 Finite receiving at the choke

At the choke, same-lane mass consumes one unit of receiving work per unit mass. Mass that still must merge consumes $\alpha > 1$ units, following the gap-acceptance receiving-work interpretation in Carey, Balijepalli, and Watling. Let $\mu > 0$ be available receiving work per hour. The complete cohort's receiving workload is

$$W_s(\tau) = c_s(\tau) + \alpha e_s(\tau) = M + (\alpha - 1) p_s M e^{-\lambda\tau}. \tag{5}$$

Neutral proportional allocation shares binding receiving between effective same-lane and gap-weighted lateral demand. For the homogeneous fluid batch used here, workload is served at rate μ, so clearance duration is

$$T_s(\tau) = \frac{W_s(\tau)}{\mu}. \tag{6}$$

The service rule is conservative and work-conserving in receiving-work units. It does not create or delete vehicle mass. The gap factor changes time/space consumption, not mass.

6 Complete affected-cohort objective

All M mass begins at the preparation-zone entrance at the decision time, traverses for τ, and then clears the choke under homogeneous proportional fluid service. Departures are linear over T_s, so

mean post-arrival time in system is $T_s/2$. The per-mass complete time score is

$$J_s(\tau) = \tau + \frac{T_s(\tau)}{2} \tag{7}$$

$$= \tau + \frac{M}{2\mu} + Kp_s e^{-\lambda\tau}, \qquad K := \frac{(\alpha-1)M}{2\mu} > 0. \tag{8}$$

This score includes the full action-induced traversal delay and the time until the complete cohort clears. No unfinished demand or upstream waiting is omitted. A common action-independent prefix cost or downstream travel time can be added without changing the comparison.

7 Parameter class and active regime

The global parameter domain is

$$M, L, \mu, \lambda > 0, \quad \alpha > 1, \quad 0 < v_{\min} < v_{\max}, \quad 0 < p_L < p_H < 1. \tag{9}$$

S13 does not select one calibrated point. The action-split certificate requires the open inequalities

$$K\lambda p_L < e^{\lambda\tau_{\min}} < K\lambda p_H < e^{\lambda\tau_{\max}}. \tag{10}$$

Strict inequalities make this a nonempty open regime whenever the common parameters permit ending-lane fractions on both sides of the threshold. They are declared before evaluating an action ranking. Equality surfaces are excluded active-set boundaries, not silently rounded cases.

8 Aggregate limits and falsification

The hidden-state contribution disappears when any of the following holds:

- $\alpha \to 1$: lateral and same-lane traffic consume equal receiving work;
- $\lambda \to 0$: advice no longer changes remaining mandatory transfer;
- $p_L = p_H$: the hidden preparations coincide; or
- receiving never binds: the workload term no longer delays clearance.

In these limits, fastest advice is common best and S05's obstruction is recovered.

The M2 claim is falsified if the exact observer collision cannot reach the zone entrance, the shared speed cannot change residence/remaining transfer, complete accounting removes the action split, or an observer-only deterministic model can condition on information that distinguishes the preparations at decision time.

9 Differentiability boundary

Inside the fixed active regime, equations (2)–(8) are smooth in continuous state and action. Lane topology and the ending-lane label are fixed metadata. Action endpoints, receiving becoming nonbinding, priority-rule changes, zero mass, and CTM demand/supply branch switches are nonsmooth or active-set boundaries. S13 makes no smoothing claim across them.

10 Source and adaptation statement

Per-lane conservation and mandatory transfer are grounded in Nagalur Subraveti, Knoop, and van Arem. Finite receiving, proportional priority, and the lane-change gap factor are grounded in Carey, Balijepalli, and Watling. Common VSL at a permanent lane drop is grounded in Zhang et al. The constant-rate active branch, batch handoff, and certificate are S13's explicit conditional analytical adaptations.